Bruno J. S. Trentini

London W9, United Kingdom British/Brazilian National

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Interests

Graphs, Machine Learning, Bayesian Methods & Nonparametrics, Causality, Evolutionary Computation, Statistical Computing.

Education

University College London — UCL United Kingdom
MSc. Data Science & Machine Learning Sep '19 – Dec '20
First Class, Distinction

University College London — UCL United Kingdom
Robotics & Computation (*Incomplete MSc.*) Sep '18 – Nov '18

Univ. Tecnologica Federal do Parana — UTFPR Brazil
PGDip. Telematics & Computer Networks Mar '12 – Sep '13
Best Monographs of the Year (10/10), GPA 0.815/1.
UK NARIC Comparability Certificate

Pontificia Univ. Catolica do Parana — PUCPR Brazil
BSc. Information Systems Jan '08 – Dec '11
BCI (CREPUQ) Scholar, GPA 72.%
UK NARIC Comparability Certificate
Summer Exchange Program at Université de Technologie de Compiègne, France (Jul '09)
Semester abroad at Bishop's University, Sherbrooke, Canada (Jan '11 - May '11)

Invited talks

Thompson Sampling at GSK [Abstract][Slides]
Nov '21 — London, United Kingdom
Optimal budget allocation in online advertisement campaigns with a scalable computational agent using Thompson Sampling and Multi-Armed Bandits techniques.

Bayesian Statistics at GSK [Slides]
Feb '21 — London, United Kingdom
Covered topics on conjugate and non-informative priors, hypothesis testing, and a drug efficiency case. Lecture given to 50 members of the data science division, including Principal Data Scientists and Apprentices at GlaxoSmithKline.

Overview of Artificial Intelligence at Econet [Slides]

Sep '19 — Curitiba, Brazil

Fundamentals of Artificial Intelligence (paradigms, different models, applications) and live coding of a vanilla neural network to technical specialists and business leaders.

Introduction to Deep Learning with GPUs

Jun '20 - Sep '20 — London, Oxford, Cambridge, United Kingdom

Spoke at multiple NVIDIA Accelerator events to startups' technical leaders and investors. Covered topics include Deep Learning, GPUs and High Performance Computing.

Industry Experience

GlaxoSmithKline (GSK)

Senior Data Scientist Jun '21 – Present — London, United Kingdom

Researching and implementing efficient and scalable models for recommender systems using Bayesian Statistics and Reinforcement Learning. Leading 5 Apprentices.

Tech: Python, SQL, Git, MS Azure.

Product Manager Oct '20 – Jun '21 — London, United Kingdom

Designed and built products for improved statistical analyses globally, reaching 1000^{+} accesses per quarter. Led advanced statistical analyses for Retail Execution optimisation and implemented ETL processes, data warehouses as well as dashboards using internal and external data. Managed 3 data engineers.

Tech: SQL, Python, MS Azure.

KIT-AR - Machine Learning Research Intern

Jun '20 – Oct '20 — London, United Kingdom

Researched Machine Learning techniques for Human Activity Recognition from multi-modal data captured with Microsoft HoloLens. Implemented supervised and unsupervised models (DBSCAN, Deep Neural Networks, Expectation Maximisation, DTW, SAX). Proposed a novel techniques for 'next-state' predictions using the Mel-Frequency Cepstrum of Sonified Spatial (3D) Data.

Tech: SQL, Python, Unity, C#, Matlab.

NVIDIA - A.I. Startups Ecosystem Manager

Dec '18 – Dec '19 — London, United Kingdom

Mentored 300^{+} startups to develop products powered by Deep Learning with NVIDIA GPUs and SDKs. Fund-raising support at seed and Series-A stage.

Tech: SQL, Python

HotelBeds Group - Commercial Analytics Manager

Oct ‘17 – Oct ‘18 — London, United Kingdom

Implemented Statistical and Machine Learning models for forecasting and anomaly detection (Holt-winters, SVM, Clustering, Decision Trees, Neural Networks). Led root-cause analyses and built several data visualisations and reports that reached 8000^+ views in 1 year. Managed 1 data scientist directly and 2 others indirectly. Contributed to reversing negative trends of 12 major accounts, which then led to a 2% YoY Revenue growth.

Tech: SQL, Python, Unity, C#, Matlab.

Alliants - Consultant

Oct ‘14 – Sep ‘17 — London, United Kingdom

Consultant for e-commerce analytics, marketing optimisation and product management for clients in the Travel and Retail industries. Conducted AB Tests, Software Tests and Statistical Analyses. Part of the team who launched the Four Seasons Mobile App.

Tech: SQL, Python, JavaScript.

Global Village Telecom - Product Manager II

Jun ‘12 – Sep ‘14 — Curitiba, Brazil

Implemented product prototypes such as a gesture-based remote control for PayTV services using Microsoft Kinect sesnors and SDK in C#. Developed products from the ground up and analysed consumer and operations data. Analysed Operations data and proposed a system configuration change that resulted in 60% savings in OpEx. Analysed Operational and Market data to support decision-making.

Tech: SQL, C#

Ernst & Young - Trainee of Advisory

Jun ‘11 – Jun ‘12 — Curitiba, Brazil

Developed scripts to detect financial misconducts, mining data from databases and identifying systemic vulnerabilities. Audited IT General Controls.

Tech: SQL, ACL, Shell

HSBC - Trainee Software Analyst

Jun ‘08 – Dec ‘10 — Curitiba, Brazil

Developed COBOL and DB2 applications in a global data orchestration and migration project running in the IBM z/OS Mainframes with teams based in China, India, UK, US and Canada.

Tech: COBOL, DB2, SQL.

Research experience

Frailty in the Very Elderly (in progress)

Mar ‘21 – Present — London, United Kingdom; Sao Paulo, Brazil;

Investigating machine learning techniques to cluster patients and to model the propensity of comorbidities in the elderly given data collected through surveys and Actimetry sensors (sleep, activity and movement). Research led by PhD. Cand. MD Fernando Blauth, University of Sao Paulo.

Kinesis to Speech: Predicting Industrial Activities Through the Mel-Frequency Cepstrum of Sonified Spatial Data [abstract]

Jun '20 – Oct '20 — London, United Kingdom

Investigated supervised and unsupervised machine learning techniques to detect, classify and predict future states of Human Activities from Microsoft HoloLens sensors' data, proposing a new algorithmic approach and a framework from data collection to inference. Developed in partnership with KIT-AR.

Advisors: Prof. Dr. Simon Julier (UCL), Dr. Felix Mannhardt (TU Eindhoven), Dr. Manuel Fradinho Oliveira (KIT-AR, SINTEF).

Breaking BERT: An Empirical Study of using BERT for Authorship Attribution [report]

Jan '20 – Mar '20 — London, United Kingdom

Comprehensive set of experiments to distill how BERT responds to several linguistic stimuli at sentence level, identifying how the model functions for text classification and a comparison against LSTM, SVM and BoW.

Advisors: Prof. Dr. Sebatien Riedl (UCL, Facebook FAIR), Dr. Tim Rocktäschel (UCL, Facebook FAIR), Dr. Pasquale Minervini (UCL). **Co-authors**: Mr. Nicolas Benielli Borrajo, Mr. Petar Hristov, Mr.Simon Lehnerer

Intelligent Transportation Systems Applied in Traffic Control [Thesis]

Mar '13 – Sep '13 — Curitiba, Brazil

Researched networks (WiMAX, WLAN, VANET, WAVE, Mesh Networks, among others) architectures and protocols and genetic algorithms. Elaborated a case study proposing a solution to traffic congestion in Curitiba, a city with 2 million inhabitants in southern Brazil.

Advisor: Prof. Dr. Walter Godoy Junior (UTF-PR)

Technical Skills

Python, Matlab, Julia, Git, SQL, SciPy, Numpy, cuDF, cuML, cuGraph, GPUs, PyTorch, TensorFlow (Keras), Scikit-Learn, JavaScript, Machine Learning, Data Science, Object Oriented Programming, Databases, TDD.

Code Samples

Odontobook [Github (Java J2EE)] Jan '11 – Dec '11

Java J2EE application intendened to connect dentists and patients, including diagnostics, prognostics and secure data sharing. Built in Java, Hybernate, JSP, Apache Tomcat Server, Oracle 10g database. **Advisors**: Dr. Marcio Fuckner, Mr. Attilio Neto.

Appied Machine Learning: Otto & Data Science Bowl Kaggle Competitions [Report] [Kernel (Python)] Jan '20 – Mar '20

Applied Machine Learning techniques for tabular and image classification (Python). **Advisor**: Dr.Dmitry Adamskiy (UCL). **Co-authors**: Mr. Conrad Whiteley, Mr. Nicolas Benielli Borrajo, Mr. Petar Hristov, Mr.Simon Lehnerer

US Census Data Analysis and Income Classification [Kernel (Python)]
Oct '20

Tutorial kernel based on the U.S. Census Data to experiment and explain basic data science & machine learning techniques.

Deep Learning Architectures on MNIST data [Kernel (Julia)] Oct '19 – Dec '19

Comparing different Neural Network architectures, optimisers and hyper-parameters on fashion-MNIST datasset (Julia). **Advisors**: Prof. Dr. Pontus Stenetorp (UCL), Dr. Pasquale Minervini (UCL), PhD Cand. Mr. Max Bartolo (UCL).

Appeatit [Github (Java)] Mar '17 – Jun '17

Android app intended to connect amateur cooks with customers in touristic zones.**Co-Author**: Mr. Tobias Guimaraes.

BugBusters [Github (Java J2ME)] Jan '08 – Dec '08

Mobile game (Runner up at the 3rd Mobile Games Expo organised by PUC-PR. Early trials of augmented reality on Symbian OS. **Co-Authors**: Mr. Andre Schuster, Mr. Luiz V. Lemberg.

Voice Classification [Github (Java)] Jun '21

Android Mobile Application to classify, given an audio data input, whether a person is singing, talking or silent using machine learning techniques.

Collection of Python, Machine Learning & Data Science Exercises [Github (Python)]
Since '17

A repository containing several exercises, experiments and learning outcomes from offline and online training since 2017.

Content Writing

Why should I always switch doors in the Monty Hall problem? [link]
March '21 — London, United Kingdom

Post proving that one should always switch doors in the classic "Monty Hall Problem". Includes bayesian perspective, visualisations and Python code.

Statistical Errors (With Aliens) [link]
January '21 — London, United Kingdom

Explanation of statistical Type-I and Type-II errors with examples.

7 Resources for Those Wanting to Learn Data Science [link]
September '18 — London, United Kingdom

Sharing my own experience in growing interest and learning data science.

Honours

Silver Award, Outstanding Contribution in the Data Science Team, GSK	2021
Bronze Award, Contributions to the de-merger activity, GSK	2021
First Class / Distinction, UCL Data Science & Machine Learning	2020
Best Business Solution at the DSS Microsoft/Amex Hackathon	2019
Invited to the London Tech Week by the UK Government	2019
Elected Department Representative, UCL Computer Science	2018
Elected Course Representative, UCL Robotics & Computation	2018
Awarded Professional Member, British Computer Society	2017
BCI (CREPUQ) Scholarship, $7,000 grant for semester abroad	2010
Selected as Representative in a Cultural Exchange, PUCPR - UTC	2009
Runner-up, 3rd Mobile Games Workshop at PUC-PR	2009

Relevant Courses

Academic

Graphical Models – UCL	2020
Applied Bayesian Methods– UCL	2020
Reinforcement Learning – UCL	2020
Statistical Natural Language Processing – UCL	2020
Introduction to Deep Learning – UCL	2020
Robotic Control Theory and Systems* – UCL	2018
Robotic Sensing, Manipulation and Interaction* – UCL	2018
Robotic Systems Engineering* – UCL	2018
Image Processing* – UCL	2018
Research Methodology – UTFPR	2013
Distributed Systems – UTFPR	2013
C & Matlab – UTFPR	2013
Data Compression & Reliability – UTFPR	2013
Distributed Systems – UTFPR	2013
Real-time Comm. Systems – UTFPR	2013
Programming I to VI – PUCPR	2008-2011
Artificial Intelligence Techniques – PUCPR	2011
General Systems Theory - PUCPR	2010
Theoretical Aspects of Computing– PUCPR	2010
Data Structures I & II– PUCPR	2009
Database Systems I & II– PUCPR	2009
Discrete Maths I & II– PUCPR	2008
Maths I & II– PUCPR	2008
Probability & Statistics I & II– PUCPR	2008

**Attended, coursework delivered, credits not obtained due to programme change.*

Professional

Building Intelligent Recommender Systems – NVIDIA	2021
Fundamentals of Accelerated Data Science with RAPIDS – NVIDIA	2021

SQL Window Functions – Vertabelo Academy	2021
Python 3.0 Bootcamp: From Zero to Hero in Python on Udemy	2018
Data Science and Machine Learning Bootcamp on Udemy	2018
Machine Learning – Coursera	2017
Introduction to Mathematical Thinking – Coursera	2017
ITIL V3 Foundation – Exin	2014

References

Prof. Dr. Simon Julier
University College London (sjulier@ucl.ac.uk)
Relationship: Master's Thesis Supervisor

Prof. Dr. Shi Zhou
University College London (s.zhou@ucl.ac.uk)
Relationship: UCL Personal Tutor

Dr. Felix Mannhardt
TU Eindhoven (f.mannhardt@tue.nl)
Relationship: Master's Thesis Co-Advisor

Dr. Manuel Fradinho Oliveira
University College London (manuel.oliveira@kit-ar.com)
Relationship: Master's Thesis Co-Advisor

Mr. Nathan Kafi
GSK (nathan.kafi@gsk.com)
Relationship: Line Manager, Principal Data Scientist

Dr. Gueorgui Mihaylov
GSK, UKRI EPSRC (gueorgui.mihaylov@gsk.com)
Relationship: Work Colleague, Principal Data Scientist

Dr. Cristina Vercosa Perez Barrios de Souza
Pontificia Universidade Catolica do Parana (cristina.souza@pucpr.br)
Relationship: Lecturer at PUC-PR (Discrete Maths)

Dr. Raquel Stasiu
Pontificia Universidade Catolica do Parana (raquel.stasiu@pucpr.br)
Relationship: Lecturer at PUC-PR (Software Engineering)

Mr. Serge Lemonde, Global A.I. Startups Director
NVIDIA (slemonde@nvidia.com)
Relationship: Line Manager at NVIDIA